



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO K.R.AGENCIES (LOCK NO -11005  
)

### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

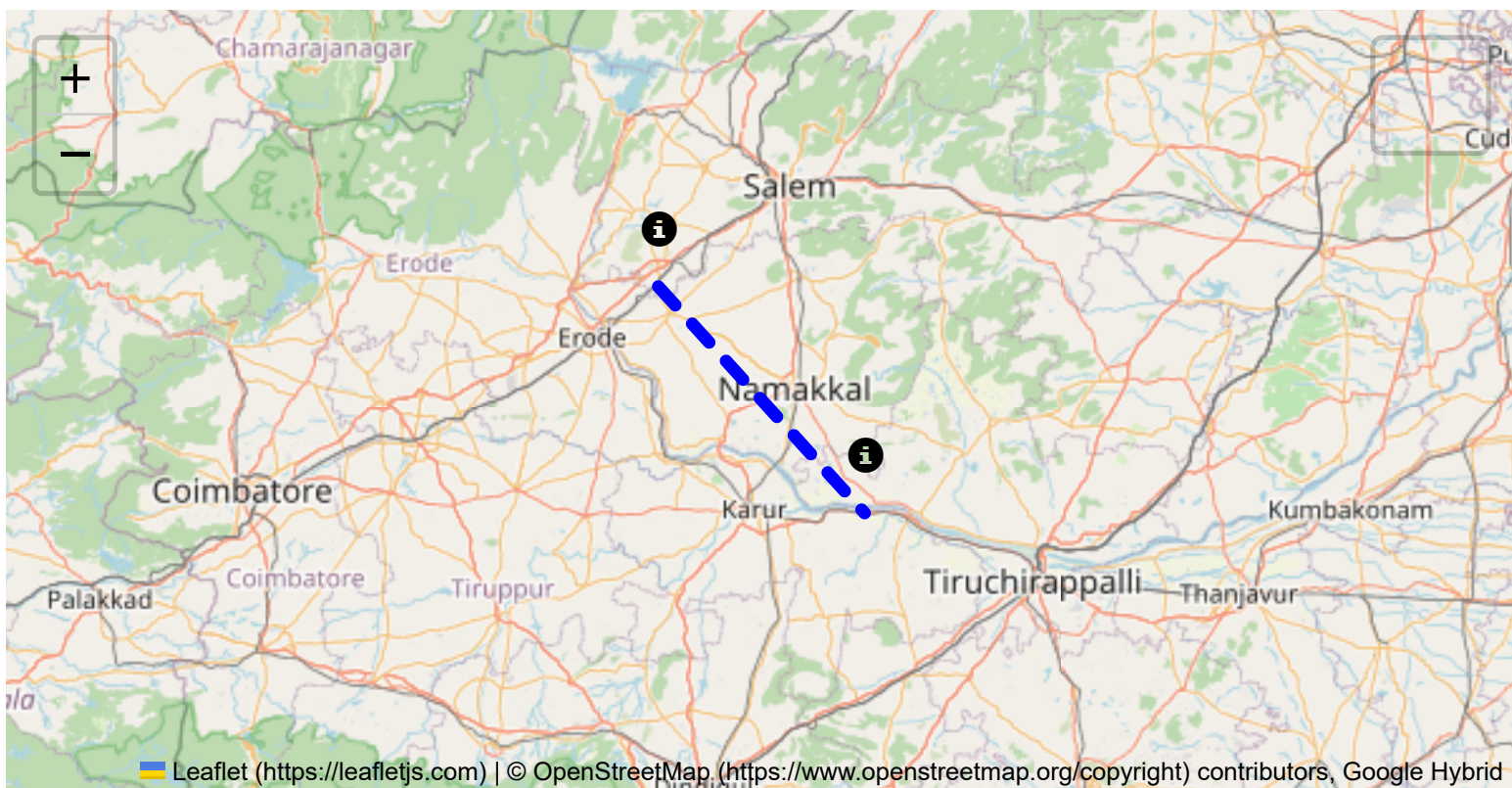
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

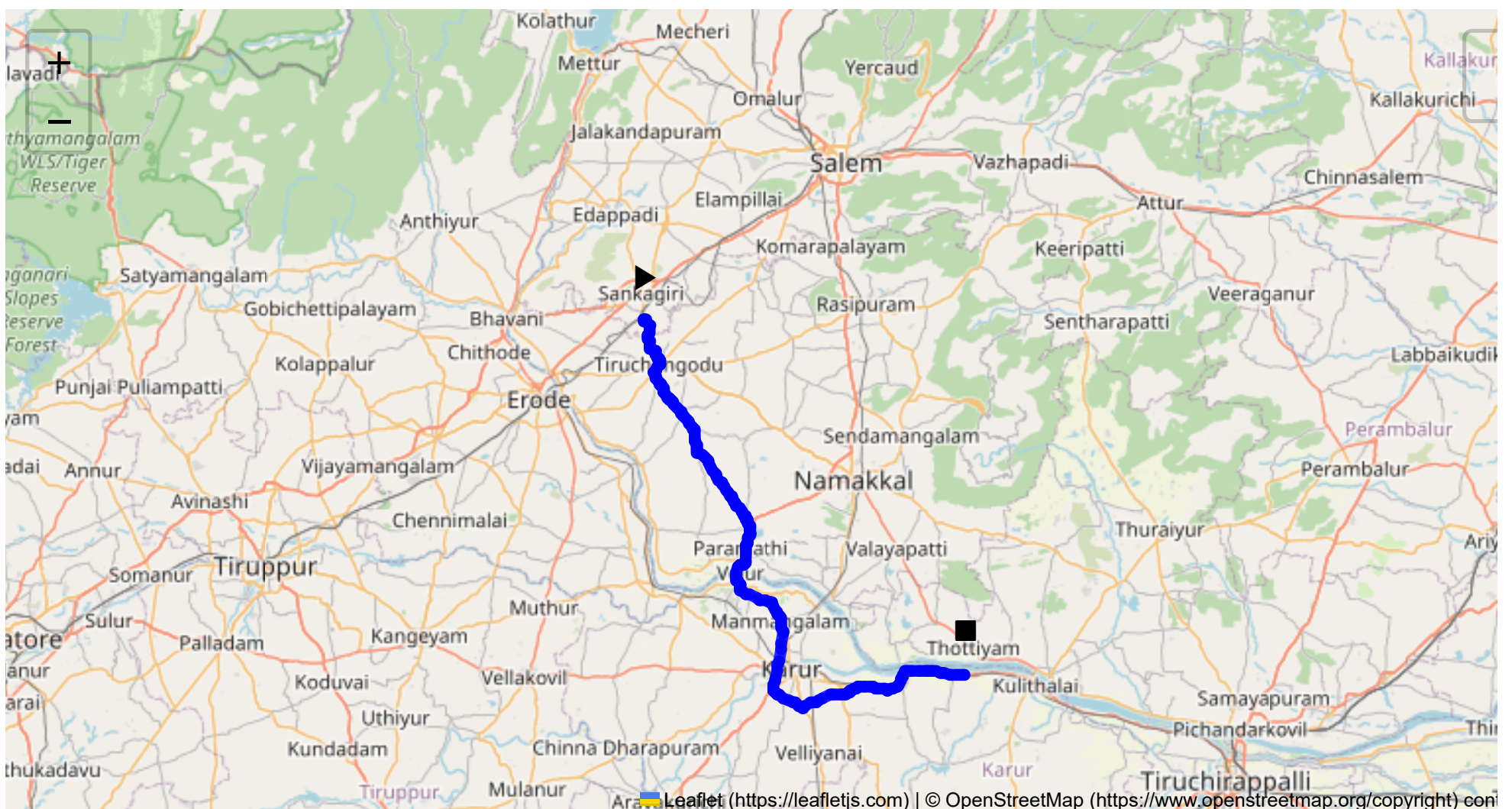
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 100.45 km**  
**Estimated Duration: 1.8 hours**  
**Adjusted Duration (Heavy Vehicle): 2.3 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.950759, 78.320975)**



## Welcome to the Journey Risk Management Study

### Route Safety Analysis: Sangagiri to Chinthalavadi

#### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to X82C+79, Chinthalavadi spans roughly 100.45 kilometers, traveling mostly through National and State Highways, including NH79. The journey marks a transition

from a more urban environment in Sangagiri to a predominantly rural area in Chinthlavadi, with several small towns and agricultural fields along the way.

## 2. Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate, characterized by hot summers, heavy monsoon rains from June to September, and mild winters. Potential weather-related hazards include:

- **Monsoon Flooding:** During peak monsoon season, roads may flood, causing delays and hazardous driving conditions.
- **Heatwaves:** In summer, heat can affect vehicle performance.

## 3. Traffic Patterns and Congestion Points

- **Peak Hours:** Typically, traffic is heavier during mornings (7-9 AM) and evenings (5-7 PM), especially near towns like Erode.
- **Congestion Areas:** Expect potential slowdowns near urban centers and local markets, particularly in Erode and Karur.

## 4. Road Quality and Infrastructure

- **National and State Highways:** These roads are generally well-maintained, but occasional potholes and uneven surfaces can be expected.
- **Rural Sections:** Beyond the main highways, roads may be narrower and less maintained, especially affecting large vehicles.

## 5. Alternative Routes for Emergencies

In case of road closures or severe traffic, consider using:

- **Alternative Route 1:** Deviate via SH 21 to connect back to NH44, offering a bypass around congested city areas.
- **Alternative Route 2:** Local district roads around the main towns can serve as interim detours.

## 6. Local Regulations on Hazardous Material Transport

Tamil Nadu authorities may impose restrictions on hazardous material transport, especially during night hours (10 PM to 6 AM) in urban areas. Permits and adherence to prescribed routes are often mandatory.

## 7. Historical Incidents

There are sporadic reports of incidents involving heavy vehicles, primarily due to mechanical failures or adverse weather conditions. Specific data on hazardous material incidents are less documented but suggest increased caution near towns.

## 8. Environmental Considerations



- **Protected Areas:** There are no significant wildlife sanctuaries along the direct route. However, agricultural zones demand caution for potential chemical spills.
- **Water Bodies:** Ensure no contamination risk to nearby rivers, especially during monsoon seasons.

## 9. Communication Coverage

- **Coverage Generally Good:** Expect adequate mobile coverage in urban and highway stretches.
- **Dead Zones:** Potential weak spots in rural areas and near expansive agricultural lands, notably away from major towns.

## 10. Emergency Response Times

- **Urban Areas:** Quick response within 20-30 minutes due to proximity to emergency services.
- **Rural Sections:** Delays up to 45 minutes to 1 hour in more remote stretches, especially those away from larger towns.

## 12. Overall Summary of Risk Assessment

- **Main Risks Identified:** Include weather-related disruptions, road quality in rural areas, and regulated transport conditions.
- **Mitigation Suggestions:** Plan ahead to avoid peak traffic hours; ensure vehicle maintenance before travel; remain vigilant during the rainy season; and have emergency contacts ready.

Overall, the route poses moderate risks mostly driven by weather and rural infrastructure conditions but can be safely navigated with cautious planning and adherence to local regulations for hazardous materials.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1 | Roundabout | High       | 10.92958, 78.05452 | 15 KM/Hr    |
| 2 | Turn       | Medium     | 11.43822, 77.87348 | 30 KM/Hr    |
| 3 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4 | Turn       | High       | 11.44019, 77.87547 | 15 KM/Hr    |
| 5 | Turn       | Medium     | 11.39685, 77.88861 | 30 KM/Hr    |
| 6 | Turn       | High       | 11.37862, 77.89488 | 15 KM/Hr    |
| 7 | Turn       | High       | 11.37850, 77.89453 | 15 KM/Hr    |
| 8 | Turn       | Medium     | 11.16764, 78.01565 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 9  | Turn      | High       | 11.14545, 78.02208 | 15 KM/Hr    |
| 10 | Turn      | High       | 11.14480, 78.02291 | 15 KM/Hr    |
| 11 | Turn      | Medium     | 11.10102, 78.00572 | 30 KM/Hr    |
| 12 | Turn      | Medium     | 10.95551, 78.23830 | 30 KM/Hr    |
| 13 | Turn      | High       | 10.95128, 78.32099 | 15 KM/Hr    |
| 14 | Turn      | High       | 10.95096, 78.32103 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                             | coordinates               | speed_limit | risk_level |
|----|----------|----------------------------------|---------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328,<br>77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331,<br>77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885,<br>77.8965774 | 30 km/h     | Medium     |
| 6  | clinic   | Kongu Nursing Home               | 11.3783065,<br>77.8961134 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429,<br>77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426        | 30 km/h     | Medium     |
| 9  | hospital | Krishna Hospital, Namakkal       | 11.3754811,<br>77.8931817 | 30 km/h     | Medium     |
| 10 | hospital | Tirukumaran Hospitals            | 11.3782118,<br>77.8914933 | 30 km/h     | Medium     |
| 13 | police   | கரட்டுப்பாளையம் காவல் நிலையம்    | 11.357586,<br>77.8922539  | 30 km/h     | Medium     |
| 14 | hospital | Goverment Hospital, Nallur       | 11.2679698,<br>77.946863  | 30 km/h     | Medium     |
| 16 | hospital | Goverment Hospital               | 11.156647,<br>78.0213586  | 30 km/h     | Medium     |
| 18 | hospital | Abishek Ortho Center             | 10.9628788,<br>78.0600715 | 30 km/h     | Medium     |
| 19 | hospital | Abishek Ortho Center, Karur      | 10.9624485,<br>78.059568  | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                                        | coordinates               | speed_limit | risk_level |
|----|-------------|-------------------------------------------------------------|---------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                                | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                               | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை                         | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution                                 | 11.3772013,<br>77.8908807 | 30 km/h     | Medium     |
| 12 | school      | MDV School                                                  | 11.3719843,<br>77.8915244 | 30 km/h     | Medium     |
| 15 | school      | Sivabakkiam Muthusamy Matriculation Higher Secondary School | 11.1788177,<br>78.0061696 | 30 km/h     | Medium     |
| 17 | school      | Goverment Higher Secondary School                           | 11.1511724,<br>78.0200723 | 30 km/h     | Medium     |
| 20 | college     | valluvar catering                                           | 10.9553473,<br>78.0606583 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480





**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.92958, 78.05452



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44019, 77.87547





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.39685, 77.88861**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.37862, 77.89488**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.37850, 77.89453**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.16764, 78.01565



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**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.10102, 78.00572



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**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.95551, 78.23830





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**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.95128, 78.32099



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.95096, 78.32103

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